Forbidden Transformations: Identity, Observation, and Lossy Compression in Distributed Epistemic Systems

Status: Working snapshot — pre-formal, under active development **Last updated:** May 2026

Abstract

This project begins with a structural rejection of observer-independent information: any fact, observation, or belief is inseparable from its provenance — the observer configuration, method, constraints, biases, and contextual embeddings that produced it. This is not a philosophical preference but a representational claim. Observations are stateful events encoding their own conditions of production, not neutral readouts of an external world.

From this foundation, several non-obvious consequences propagate through constraint.

On context as reusable topology. If observations encode their origins, the recursive expansion of provenance across belief networks threatens intractability. This resolves partially through recognition that context possesses factorizable structure: nodes common across a given frame may collapse without loss within that frame, introducing natural compression into epistemic graph structure. Context is not merely background — it is shared infrastructure with reusable topology.

On convergence as compression. When two distinct representations converge, dimensionality is reduced and structure is necessarily lost. Because reality is not losslessly compressible, all convergence alters the epistemic object it claims to represent. Agreement between observers is not additive or neutral — it is a compression event with associated information cost.

On identity as transformation geometry. The framework's most formally tractable claim: a system's identity is more precisely characterized by its *forbidden transformations* than by its current instantiation. Identity is not a snapshot of present state but the geometry of admissible change — the manifold of possible transformations a system can undergo while remaining itself. This maps naturally onto topological and category-theoretic structures, where identity is represented less as node attributes and more as constrained morphisms. Static essence gives way to dynamical continuity across bounded transformation space.

On perspective as active state. Observation alters topology; observers are embedded within systems they observe; no detached global frame is available. The observer-observation boundary is a lossy compression artifact, not a clean partition. Any direct observation is either self-inclusive or incomplete — observer omission is not neutral abstraction but a specific form of information loss.

Taken together, these claims converge toward a unified structure: identity, observation, provenance, convergence, and cognition are variations of *stateful constrained transformations*

across distributed observers operating under finite representational capacity. Convergence is compression. Divergence is preserved dimensionality. Identity is admissible transformation geometry. Cognition is distributed topology management under compression constraint.

Formal Territory

The framework is currently pre-formal but not pre-formalizable. The structures it identifies already possess mathematical machinery in adjacent disciplines:

- **Topological data analysis** — persistence, shape of data across scales
- **Dynamical systems theory** — attractors, stability, phase transitions, bifurcation
- **Category theory** — morphisms, functors, admissible transformations as identity
- **Active inference / free energy principle** — observer-coupled state spaces, prediction under constraint
- **Information theory** — compression bounds, lossy encoding, channel capacity
- **Epistemic logic and formal epistemology** — belief revision, provenance tracking, contextualized knowledge

The near-term project is to cash these intuitions out as explicit models: toy formalizations, computable invariants, distinguishing predictions, and domain-specific instantiations that constrain rather than merely describe.

Relevance

The framework's immediate application space is sharpened by contemporary AI development, which is forcing multiple disciplines to simultaneously confront:

- representation limits and contextual collapse,
- observer-state effects on measurement and inference,
- lossy compression as epistemic distortion,
- and distributed cognition under finite capacity.

These are precisely the phenomena the framework addresses from a generalized, domain-agnostic perspective. The underlying machinery is not discipline-specific — it describes a class of system behavior that appears across cognitive science, machine learning, social epistemology, and complex systems.

A secondary application: institutions that claim observer-independent evaluation while operating through highly stateful trust networks are themselves subject to the framework's analysis. Credentialing, peer review, and expertise hierarchies are identity-coupled epistemic

filters — not neutral mechanisms, but provenance-weighted inference systems with embedded convergence pressures. This is not a critique but a description.

Anti-Mystical Commitment

This project advances claims only insofar as they admit:

- operational definition,
- formal constraint,
- eventual empirical or computational contact,
- and falsifiable or distinguishing predictions.

Where language becomes universalist, detached from computable structure, or irrefutable by construction, it has failed — regardless of how coherent it feels. The test is not internal consistency but external constraint: does the framework compress complexity better than existing models, preserve important structure, generate useful predictions, or produce computable representations?

That standard is explicit, non-negotiable, and self-applying.

Open Questions / Active Threads

- Can "forbidden transformation" identity be formalized as a computable constraint set for a non-trivial domain?
- What is the distinguishing prediction of the provenance-weighted inference model versus standard Bayesian updating?
- How does frame-local context collapse interact with cross-frame translation — what is preserved, what is lost?
- Is there a natural measure on "compression cost" of social convergence events?
- What is the relationship between observer embedding and the frame-dependence problems in quantum measurement?

Working document. Arguments are stated as claims pending formalization, not as established results.